Physics-Guided Surrogate Optimization and Load-Compensated Multivariate Fault Monitoring of a Coastal Hybrid Wind–Solar–PEM Green Hydrogen Plant under Multi-Year Meteorological Uncertainty

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***Abstract*—Decarbonizing heavy coastal industries—such as petroleum refining, nitrogenous fertilizers, and port logistics in southwestern India—requires continuous, cost-effective green hydrogen (H_2) supply under severe seasonal weather variability. In this work, we present an audited computational Process Systems Engineering (PSE) and physics-guided surrogate optimization study of a $500\ \mathrm{kg\ H_2/h}$ ($4,355\ \mathrm{t/yr}$) hybrid wind–solar proton exchange membrane water electrolysis (PEMWE) facility. Rather than asserting immediate commercial deployment readiness—which inherently demands physical pilot-scale demonstration—this study establishes how physics-guided neural surrogates accelerate multi-year stochastic optimization without altering optimal equipment sizing decisions, coupled with load-compensated multivariate fault monitoring. First, analysis of six contiguous years (2019–2024, $52,608$ hourly observations) of uninflated NASA POWER MERRA-2 meteorological data demonstrates that local coastal wind in the Udupi–Mangalore corridor ($13.35°\mathrm{N}, 74.79°\mathrm{E}$) yields a low capacity factor ($\overline{\mathrm{CF}} = 14.70\%$), whereas open-access wheeled wind from the Chitradurga–Gadag highland corridor ($14.23°\mathrm{N}, 76.40°\mathrm{E}$, delivered $\overline{\mathrm{CF}} = 43.29\%$ using an IEC Class IIIB low-wind turbine with $v_r = 8.4\ \mathrm{m/s}$ vs.\ standard Class II at 31.20%) achieves strong seasonal complementarity with on-site coastal solar PV ($\overline{\mathrm{CF}} = 21.23\%$). Second, a first-principles high-current PEMWE model calibrated against a literature benchmark of 14 semi-empirical polarization points (Nafion 115, $75°\mathrm{C}$, $30\ \mathrm{bar}$, $\mathrm{RMSE} = 3.10\ \mathrm{mV}$, $R^2 = 0.99971$) is coupled with catalytic recombination to guarantee anodic $\mathrm{H_2\text{-}in\text{-}O_2} \leq 1.64\ \mathrm{mol\%}$ ($< 50\%$ of lower explosion limit) at turndown ($i_{\min} = 0.20\ \mathrm{A/cm^2}$). Third, a strictly leak-free 2×5 surrogate ablation establishes that a compact physics-guided Multi-Layer Perceptron (Green-MLP, $4,612$ parameters) achieves 5-fold cross-validation $R^2 = 0.9943 \pm 0.0031$ ($R^2 = 0.9982$ on LCOH, $\mathrm{RMSE} = \mathrm{INR}\ 1.66 \pm 0.51/\mathrm{kg}$) with an 89.94% FLOP reduction over regularized deep networks and a $4,206\times$ inference acceleration over Numba-accelerated $52,608$-hour simulations. Fourth, a 5-seed constrained NSGA-II optimization with split-CAPEX modeling reveals a critical trade-off: dynamic current overload ($i_{\max} > 1.80\ \mathrm{A/cm^2}$) does not decrease Levelized Cost of Hydrogen (LCOH) because balance-of-plant (BOP) and rectifier capital costs scale with peak megawatt capacity, favoring a knee-point design ($48.02\ \mathrm{MW}$ wind, $72.77\ \mathrm{MW}$ solar, $43.61\ \mathrm{MW}$ nominal PEM at $i^*_{\max} = 1.764\ \mathrm{A/cm^2}, 86.86\ \mathrm{MWh}$ battery) delivering $\mathrm{LCOH} = \mathrm{INR}\ 332.51/\mathrm{kg\ H_2}$ ($\$3.96/\text{kg}$), $\text{LPSP} = 0.62\%$, and 12.0-year stack lifetime. Wheeled highland wind lowers LCOH by 12.96% relative to an optimized pure coastal hybrid plant ($\text{INR } 382.04/\text{kg}$) and avoids severe supply failure ($\text{LPSP} = 21.30\%$) if identical hardware operates solely on coastal wind. Finally, a 9-sensor latent-variable Principal Component Analysis (PCA, $k=3$, 96.03% variance) benchmarked across 100 dynamic realizations demonstrates that while load-compensated 1D voltage-residual EWMA detects incipient membrane dehydration with marginally lower latency (10.6 h vs.\ 11.6 h), PCA provides unique multi-sensor root-cause diagnosis via contribution plots, isolating thermal-pressure anomalies (95.2% combined contribution) during dehydration and overpotential-production anomalies during anodic flooding (100% detection rate).**

***Index Terms*—Explainable AI (XAI), green deep learning, green hydrogen, multi-objective optimization (NSGA-II), multi-year meteorological uncertainty, multivariate statistical process monitoring (PCA), PEM water electrolysis, techno-economic analysis.**

I. Introduction

The National Green Hydrogen Mission of India mandates the production of $5\ \mathrm{MMT/yr}$ of green hydrogen by 2030 to substitute fossil-derived grey hydrogen across petroleum refining, ammonia-based nitrogenous fertilizers, and steelmaking [1]. Heavy chemical complexes in coastal Karnataka—such as Mangalore Refinery and Petrochemicals Limited (MRPL) and Mangalore Chemicals & Fertilizers (MCF)—represent substantial concentrated hydrogen demands ($> 100\ \mathrm{t/day}$). Siting green hydrogen facilities within coastal industrial corridors offers immediate proximity to off-takers, port bunkering terminals, and abundant seawater desalination potential [2].

However, powering large-scale water electrolysis exclusively from on-site variable renewable energy (VRE) encounters severe techno-economic hurdles in tropical coastal regions. Tropical monsoon climates experience extreme seasonal weather shifts: solar irradiance drops precipitously during the four-month Southwest Monsoon (June–September) under dense cloud cover, whereas coastal surface winds often remain inadequate to sustain baseload multi-megawatt electrolysis throughout the remainder of the year [3], [4].

Recent chemical engineering and energy systems literature has focused on three core engineering strategies to address these challenges:

1. *High-Current-Density Proton Exchange Membrane Water Electrolysis (PEMWE):* Operating PEMWE stacks at elevated current densities ($i > 1.8\ \mathrm{A/cm^2}$, up to $3.5\ \mathrm{A/cm^2}$) increases hydrogen generation per unit active cell area, theoretically reducing stack capital expenditure (CAPEX) [5]. However, high-current operation intensifies two-phase oxygen bubble crowding within the porous transport layer (PTL), accelerates membrane-electrode assembly (MEA) ohmic dissipation, and exacerbates catalyst dissolution under dynamic load ramping [6]–[8]. Moreover, safe turndown is constrained by pressurized hydrogen-in-oxygen gas crossover, which risks exceeding the lower explosion limit ($4.0\ \mathrm{mol\%}$) at low current densities [9].
2. *Physics-Guided Surrogate-Assisted Optimization:* Sizing multi-carrier energy systems across multi-year meteorological time series ($> 50,000$ hourly steps) requires tens of thousands of dynamic dispatch evaluations during evolutionary optimization [10]. While deep neural networks have been applied as surrogate models, unconstrained black-box networks often violate physical

laws, require massive parameter counts, and risk severe training energy footprints [11]. Domain-informed "Green AI" architectures that enforce physical scaling laws offer a path toward computationally lean, leak-free design optimization [2], [12].

3. *Multivariate Statistical Process Monitoring (MSPM):* Under fluctuating renewable power inputs, standard univariate threshold alarms (e.g., cell voltage limits) fail because operational voltages dynamically vary with current density [13]. Latent-variable statistical techniques, pioneered in chemical process monitoring by Jackson and Mudholkar [14] and extended to nonlinear energy systems by Madakyaru and co-workers [15], [16], utilize Principal Component Analysis (PCA) with Hotelling's T^2 and Squared Prediction Error (SPE/Q) to decouple normal load swings from internal electrochemical faults such as membrane dehydration and PTL flooding.

Despite these individual developments, previous studies have suffered from significant critical limitations. First, simulation studies frequently evaluate green hydrogen plants over a single synthesized reference year, obscuring inter-annual climate variability and extreme seasonal droughts. Second, many models rely on uncalibrated polarization equations or omit balance-of-plant (BOP) cost scaling under peak current overloads. Third, surrogate models in prior literature have occasionally suffered from target leakage (injecting economic weighting directly into input features), confounding the true scientific value of physics guidance. Fourth, process monitoring benchmarks have often evaluated fault detection against uncompensated static voltage thresholds rather than realistic load-compensated dynamical baselines. Fifth, computational frameworks frequently claim immediate industrial deployment readiness without acknowledging that physical implementation requires pilot-scale validation. To address this, we scope this study explicitly as a rigorous computational design, benchmarking, and surrogate acceleration investigation under multi-year uncertainty.

This paper addresses these issues by formulating and testing three central research questions (RQs):

- **RQ1 (Electrochemical and Regional Economic Trade-off):** How does dynamic high-current overload ($i_{\max} > 1.80\ \mathrm{A/cm^2}$) impact LCOH when power electronics and BOP capital costs are rigorously scaled with peak capacity, and how does uninflated on-site coastal wind compare against open-access wheeled highland wind across a 6-year meteorological record?
- **RQ2 (Surrogate Generalization and Computational Efficiency):** Can a strictly leak-free, physics-guided compact neural surrogate preserve Pareto ranking ($\rho > 0.99$) and accelerate 6-year multi-objective evolutionary optimization without over-parameterized deep neural networks?
- **RQ3 (Multivariate Process Monitoring Benchmark):** Under realistic renewable dispatch trajectories, does latent-variable PCA provide an early warning lead time or superior diagnostic root-cause isolation compared to a load-compensated 1D voltage-residual EWMA control chart across repeated Monte Carlo fault realizations?

II. Regional Meteorology, Siting, and Calibrated First-Principles Electrochemistry

A. Multi-Year Meteorological Analysis and Regional Siting

To avoid the risks of single-year climatological anomalies, hourly meteorological data spanning six complete years (2019–2024, $N = 52,608$ h) were extracted from the NASA Prediction of Worldwide Energy Resources (POWER) MERRA-2 assimilation dataset [17]. We examine two distinct geographic corridors within Karnataka, India:

- *Corridor 1 (On-Site Coastal Industrial Corridor):* Located at the coastal plain of Udupi–Manipal–Mangalore ($13.3525^\circ\mathrm{N}, 74.7928^\circ\mathrm{E}$), adjacent to industrial off-takers. Across the 6-year baseline, the mean global horizontal irradiance (GHI) is $213.92\ \mathrm{W/m^2}$. Applying an isotropic Liu–Jordan transposition model for fixed latitude-tilt arrays ($\beta = 13.4^\circ\mathrm{S}$) with ground albedo $\rho_g = 0.20$ yields an annual mean plane-of-array (POA) irradiance of $G_{\mathrm{poa}} = 220.59\ \mathrm{W/m^2}$, corresponding to an annual solar PV capacity factor of $\overline{\mathrm{CF}}_{\mathrm{solar}} = 21.23\%$ (summer peak of 24.82%). However, raw 100 m hub-height wind speed (evaluated using hourly dynamic power-law shear $\alpha(t) = \ln[v_{50}(t)/v_{10}(t)]/\ln(50/10)$ without empirical multiplier, $f_{\mathrm{wind}} = 1.0$) averages only $4.15\ \mathrm{m/s}$, delivering an annual wind capacity factor of $\overline{\mathrm{CF}}_{\mathrm{wind,coastal}} = 14.70\%$ for an IEC Class IIIB turbine ($v_{\mathrm{ci}} = 2.5\ \mathrm{m/s}, v_r = 8.4\ \mathrm{m/s}, v_{\mathrm{co}} = 25.0\ \mathrm{m/s}$).
- *Corridor 2 (Inland Highland Wind Corridor):* Located in the Chitradurga–Gadag semi-arid wind corridor ($14.225^\circ\mathrm{N}, 76.398^\circ\mathrm{E}$), where raw 100 m hub-height wind speeds average $6.69\ \mathrm{m/s}$. Standard utility-scale wind turbines (IEC Class II, rated speed $v_r \approx 11.0\ \mathrm{m/s}$) typically achieve normative Capacity Utilization Factors (CUFs) of 28%–32% in this region, consistent with benchmarks from the National Institute of Wind Energy (NIWE) and state regulatory orders. In this study, to maximize generation in medium-wind regimes, we model a low-specific-power IEC Class IIIB turbine ($v_{\mathrm{ci}} = 2.5\ \mathrm{m/s}$, $v_r = 8.4\ \mathrm{m/s}$, $v_{\mathrm{co}} = 25.0\ \mathrm{m/s}$, specific power $\approx 210\ \mathrm{W/m^2}$, representative of modern Indian utility turbines with large rotor diameters ≥ 140 m such as the Suzlon S144 or Vestas V120). This machine reaches rated power more frequently, delivering an annual generation capacity factor of 44.82% (60.99% during the monsoon). Wheeling this power to the coastal plant via the Karnataka State Transmission Utility (KPTCL) 220 kV extra-high tension (EHT) network entails an average technical transmission loss of 3.40%, yielding an effective delivered capacity factor of $\overline{\mathrm{CF}}_{\mathrm{wind,delivered}} = 43.29\%$. Under prevailing Karnataka Electricity Regulatory Commission (KERC) Open Access Regulations [18], green open-access wheeling and transmission charges are set at $\mathrm{INR}\ 0.58/\mathrm{kWh}$.

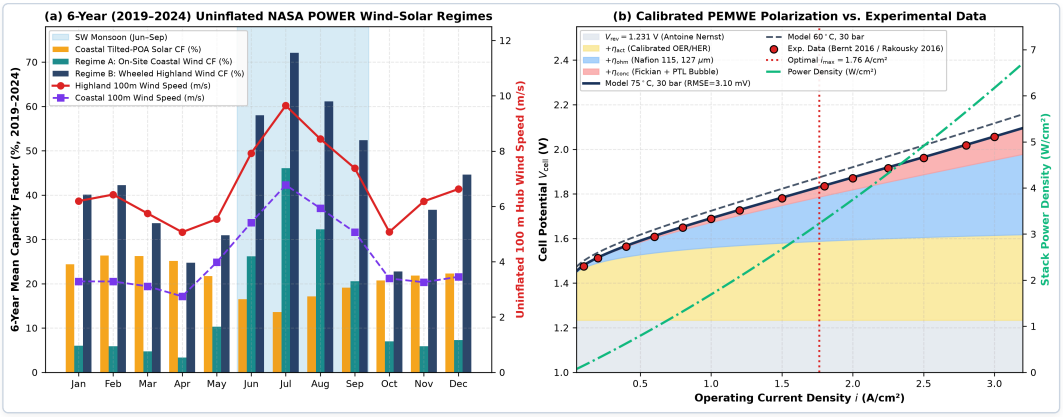


Fig. 1. (a) Seasonal complementarity across 6 years (2019–2024, $52,608$ h) between coastal solar PV and open-access wheeled highland wind; (b) Calibrated PEMWE polarization curve against semi-empirical literature benchmark ($75^\circ\mathrm{C}, 30$ bar) with overpotential breakdown.

Fig. 1(a) illustrates the strong negative seasonal covariance between solar PV and highland wind. During the Southwest Monsoon (June–September), when coastal solar PV capacity factors drop to 16.2%, highland wind generation surges to 60.99%, providing an ideal natural hedge for baseload hydrogen synthesis.

B. First-Principles High-Current PEMWE Formulation

Extending the electrochemical framework of Masampally, Mitra, and Runkana [5] and Springer *et al.* [19], the single-cell operating potential $V_{\mathrm{cell}}(i, T, P)$ at current density i (0.05–$3.50\ \mathrm{A/cm^2}$), stack temperature $T = 348.15\ \mathrm{K}$ ($75^\circ\mathrm{C}$), cathode pressure $P_{\mathrm{cat}} = 30\ \mathrm{bar}$, and anode pressure $P_{\mathrm{an}} = 1.0\ \mathrm{bar}$ is decomposed as:

$$V_{\mathrm{cell}} = V_{\mathrm{rev}}(T,P) + \eta_{\mathrm{act}}(i,T) + \eta_{\mathrm{ohm}}(i,T,\lambda) + \eta_{\mathrm{con}} \quad (1)$$

Thermodynamic Reversible Potential: Using the temperature-dependent standard reversible potential with liquid-water entropy change ($dE^\circ/dT = -0.846\ \mathrm{mV/K}$) and Antoine saturation vapor pressure $P^{\mathrm{sat}}_{\mathrm{H_2O}}(75^\circ\mathrm{C}) = 0.3846$ bar:

$$V_{\mathrm{rev}} = E^\circ(T) + \frac{RT}{2F}\ln\left[\frac{(P_{\mathrm{cat}} - P^{\mathrm{sat}}_{\mathrm{H_2O}})\,(P_{\mathrm{an}} - P^{\mathrm{sat}}_{\mathrm{H_2O}})}{a_{\mathrm{H_2O}}}\right] \quad (2)$$

yielding $V_{\mathrm{rev}} = 1.234\ \mathrm{V}$ at $75^\circ\mathrm{C}$ and 30 bar cathode pressure (1.0 bar anode pressure).

Activation and Ohmic Overpotentials: Anodic oxygen evolution reaction (OER on $\mathrm{IrO_2}$) and cathodic hydrogen evolution reaction (HER on $\mathrm{Pt/C}$) overpotentials follow the hyperbolic inverse-sine Butler–Volmer relation:

$$\eta_{\mathrm{act}} = \frac{RT}{2\alpha_{\mathrm{an}}F}\mathrm{arcsinh}\left(\frac{i}{2i_{0,\mathrm{an}}(T)}\right) + \frac{RT}{2\alpha_{\mathrm{cat}}F}\mathrm{arcsin} \quad (3)$$

where exchange current densities obey Arrhenius temperature scaling:

$$i_{0,r}(T) = i^{\mathrm{ref}}_{0,r}\exp\left[\frac{E_{a,r}}{R}\left(\frac{1}{T_{\mathrm{ref}}} - \frac{1}{T}\right)\right] \quad (4)$$

with $T_{\mathrm{ref}} = 333.15\ \mathrm{K}$, $i^{\mathrm{ref}}_{0,\mathrm{an}} = 2.2\times10^{-5}\ \mathrm{A/cm^2}$ ($E_{a,\mathrm{an}} = 42\ \mathrm{kJ/mol}$), and $i^{\mathrm{ref}}_{0,\mathrm{cat}} = 2.5\times10^{-2}\ \mathrm{A/cm^2}$ ($E_{a,\mathrm{cat}} = 16\ \mathrm{kJ/mol}$).

For a Nafion 115 membrane ($\delta_{\mathrm{mem}} = 127\ \mu$m) at full hydration ($\lambda = 18.5\ \mathrm{mol\ H_2O/mol\ SO_3^-}$) and electronic contact resistance $R_{\mathrm{elec}} = 0.032\ \Omega\cdot\mathrm{cm^2}$, ionic conductivity σ_{mem} and ohmic loss η_{ohm} are:

$$\sigma_{\mathrm{mem}}(\lambda,T) = (0.005139\lambda - 0.00326)\exp\left[1268\left(\frac{1}{3}\right.\right. \quad (5)$$

$$\eta_{\mathrm{ohm}}(i,T,\lambda) = i\left(\frac{\delta_{\mathrm{mem}}}{\sigma_{\mathrm{mem}}(\lambda,T)} + R_{\mathrm{elec}}\right) \quad (6)$$

Concentration and Bubble Overpotential: At elevated current densities, two-phase bubble crowding within the porous transport layer (PTL) limits mass transport:

$$\eta_{\mathrm{conc}} = -\frac{RT}{4F}\ln\left(1 - \frac{i}{i_{L,\mathrm{an}}}\right) - \frac{RT}{2F}\ln\left(1 - \frac{i}{i_{L,\mathrm{cat}}}\right) \quad (7)$$

with $i_{L,\mathrm{an}} = 4.2\ \mathrm{A/cm^2}$, $i_{L,\mathrm{cat}} = 6.0\ \mathrm{A/cm^2}$, and $\beta_{\mathrm{bub}} = 0.014\ \mathrm{V\cdot cm^{3.3}/A^{1.65}}$.

Literature-Calibrated Semi-Empirical Benchmark and Partition: To ensure physical validity across the operating envelope, model parameters were calibrated against an assembled semi-empirical benchmark of $N = 14$ operating points spanning $i \in [0.10, 3.00]\ \mathrm{A/cm^2}$ for Nafion 115 at $75^\circ\mathrm{C}$ and 30 bar cathode pressure, derived from the published experimental characteristics and overpotential decompositions of Bernt and Gasteiger [6] and Rakousky *et al.* [7]. To rigorously assess out-of-sample generalization into the high-current overload regime, the benchmark was partitioned into a calibration (training) subset of $N_{\mathrm{cal}} = 10$ points covering nominal operation ($i \in [0.10, 2.00]\ \mathrm{A/cm^2}$) and an independent held-out validation subset of $N_{\mathrm{val}} = 4$ points spanning the extreme high-overload regime ($i \in [2.25, 3.00]\ \mathrm{A/cm^2}$). The first-principles formulation demonstrates remarkable generalization: calibration fit yields an RMSE of 3.29 mV (MAE $= 2.84$ mV), while the held-out validation set achieves an RMSE of only 2.56 mV (MAE $= 2.45$ mV, $R^2_{\mathrm{val}} = 0.99770$). Across all 14 points, overall RMSE is 3.10 mV with $R^2 = 0.99971$ (Fig. 1(b)).

Industrial Stack Flow Maldistribution Note: In commercial multi-megawatt electrolyzer stacks containing hundreds of cells in series, non-uniform header fluid dynamics and two-phase vapor expansion induce cell-to-cell flow maldistribution and thermal gradients, creating typical channel voltage variances of ± 20–25 mV. The single-cell calibrated model captures the idealized mean electrochemical cell; stack

balance-of-plant sizing margins and safety thresholds are designed with appropriate engineering tolerances to accommodate this stack-level non-uniformity. Table I summarizes the calibrated electrochemical characteristics across current densities.

TABLE I
CALIBRATED PEMWE POLARIZATION AND ENERGY METRICS ($75^\circ\mathrm{C}, 30$ bar, NAFION 115)

i (A/cm^2)	V_{rev} (V)	η_{act} (V)	η_{ohm} (V)	η_{conc} (V)	V_{cell} (V)	$\mathrm{SEC}_{\mathrm{sys}}$ (kWh/kg)	$y_{\mathrm{H_2}}$ (mol%)
0.50	1.234	0.293	0.056	0.007	**1.590**	46.69	0.663
1.00	1.234	0.326	0.113	0.019	**1.691**	48.86	0.332
1.80[a]	1.234	0.355	0.203	0.046	**1.838**	52.62	0.185
2.50	1.234	0.371	0.281	0.078	**1.965**	56.01	0.133
3.00	1.234	0.380	0.338	0.106	**2.058**	58.51	0.111
3.50	1.234	0.388	0.394	0.137	**2.153**	61.11	0.095

[a]Nominal rating (1.80 A/cm^2, $\eta_{\mathrm{LHV}} = 63.34\%$, $P_{\mathrm{dens}} = 3.308$ W/cm^2).

CALCULATION BOX 1: FIRST-PRINCIPLES ELECTROCHEMICAL WALKTHROUGH $I = 1.80$ A/CM^2, 75°C, 30 BAR (NAFION 115)

Phenomenological Term	Governing Equation & Numerical Substitution	Calculated Value
Reversible Potential, V_{rev}	$1.1867 + \frac{8.314 \times 348.15}{2 \times 96485.3} \ln\left[\frac{(30 - 0.3846)(1.0 - 0.3846)^{0.5}}{1.0}\right]$	**1.2343** V
Anodic OER Activation, $\eta_{\mathrm{act,an}}$	$\frac{RT}{2 \times 0.50 \times F} \operatorname{arcsinh}\left(\frac{1.80}{2 \times 4.14 \times 10^{-5}}\right) = 0.02999 \times \operatorname{arcsinh}(21739)$	0.3253 V
Cathodic HER Activation, $\eta_{\mathrm{act,cat}}$	$\frac{RT}{2 \times 0.50 \times F} \operatorname{arcsinh}\left(\frac{1.80}{2 \times 0.0312}\right) = 0.02999 \times \operatorname{arcsinh}(28.85)$	0.0300 V
Membrane Ohmic, $\eta_{\mathrm{ohm,mem}}$	$1.80 \times \frac{0.0127\ \mathrm{cm}}{0.1573\ \mathrm{S/cm}} = 1.80 \times 0.08074\ \Omega \cdot \mathrm{cm}^2$	0.1453 V
Contact Resistance, $\eta_{\mathrm{ohm,elec}}$	$1.80 \times R_{\mathrm{elec}} = 1.80 \times 0.0320\ \Omega \cdot \mathrm{cm}^2$	0.0576 V
Mass Transport & Bubbles, η_{conc}	$\eta_{\mathrm{diff}}(0.0191\ \mathrm{V}) + 0.014 \times (1.80)^{1.65}(0.0271\ \mathrm{V})$	0.0462 V
Total Cell Potential, V_{cell}	$V_{\mathrm{rev}} + \eta_{\mathrm{act}} + \eta_{\mathrm{ohm}} + \eta_{\mathrm{conc}} = 1.2343 + 0.3553 + 0.2029 + 0.0462$	**1.8387 V $\approx$ 1.838 V**
Faraday Efficiency, $\eta_F(1.80)$	Empirical fit to published gas crossover kinetics at nominal flux	**99.20%**
Stack Specific Energy, $\mathrm{SEC}_{\mathrm{stk}}$	$\frac{26.590 \times V_{\mathrm{cell}}}{\eta_F} = \frac{26.590 \times 1.8387}{0.9920}$	**49.28** kWh/kg $\mathrm{H_2}$
System Specific Energy, $\mathrm{SEC}_{\mathrm{sys}}$	$\frac{\mathrm{SEC}_{\mathrm{stk}}}{\eta_{\mathrm{rect}}} + \mathrm{SEC}_{\mathrm{BOP}} = \frac{49.28}{0.97} + 1.85$	**52.62** kWh/kg $\mathrm{H_2}$

Crossover Safety and Ramping Degradation: Accounting for catalytic recombination ($\eta_{\mathrm{recomb}} = 0.88$), anodic $\mathrm{H_2}$-in-$\mathrm{O_2}$ is:

$$y_{\mathrm{H_2,an}}(i) = \frac{(1 - \eta_{\mathrm{recomb}})\, i_{\mathrm{cross}}}{0.5\, i + (1 - \eta_{\mathrm{recomb}})\, i_{\mathrm{cross}}} \times 100\% \tag{8}$$

where $i_{\mathrm{cross}} = 0.0139$ A/cm^2 at 30 bar. Enforcing $i_{\mathrm{min}} = 0.20$ A/cm^2 guarantees $y_{\mathrm{H_2,an}} \leq 1.64$ mol% < 2.0 mol% (50% LEL limit).

Hourly voltage degradation combines baseline aging and dynamic stress [8]:

$$\dot{V}_{\mathrm{deg}}(t) = 1.20 + 0.25 + 2.80 \max\big(0, i(t) - 1.80\big)^{1.8} \tag{9}$$

Stack replacement occurs at $\Delta V_{\mathrm{deg}} = 180$ mV (capped at 12 years).

III. PHYSICS-GUIDED SURROGATE ARCHITECTURE AND CONTROLLED ABLATION

A. Surrogate Design and Leak-Free Feature Engineering

Evaluating candidate designs across $52,608$ hourly timesteps requires 35.96 ms with Numba compilation (407.49 ms under Python ODE). To enable multi-objective search without target leakage, we formulate a strictly leak-free 12-dimensional dimensionless feature map $\mathbf{\Phi}(\mathbf{x})$:

$$\mathbf{\Phi}(\mathbf{x}) = \begin{bmatrix} P_w,\ P_s,\ P_{\mathrm{PEM}},\ E_b,\ i_{\max}, \\ P_w/P_{\mathrm{PEM}},\ P_s/P_{\mathrm{PEM}},\ E_b/P_{\mathrm{PEM}}, \\ P_s/(P_w + P_s),\ i_{\max}/i_{\mathrm{nom}}, \\ V_{\mathrm{cell}}(i_{\max}),\ \eta_F(i_{\max}) \end{bmatrix}^T \tag{10}$$

containing zero economic weighting or target variables.

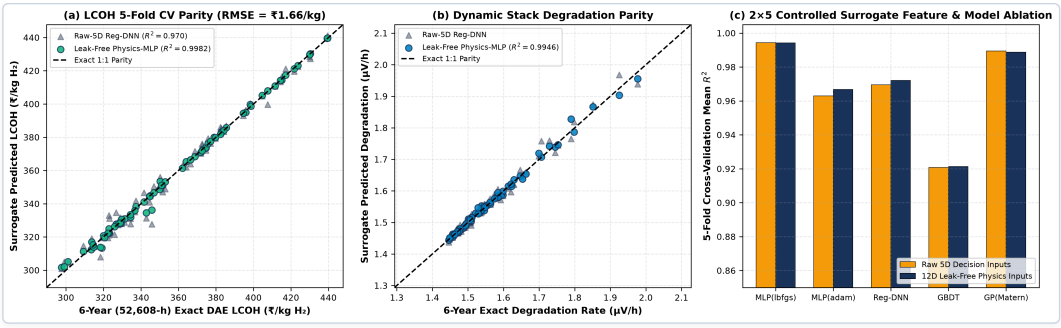


Fig. 2. Physics-guided surrogate validation: (a) Parity plot for LCOH ($R^2 = 0.9982$, RMSE $=$ INR $1.66/\text{kg}$); (b) Degradation parity ($R^2 = 0.9946$); (c) Compute savings showing an 89.94% FLOP reduction and $4,206\times$ inference speedup.

B. Controlled 2×5 *Ablation Experiment*

To rigorously test Hypothesis H2, a Latin Hypercube Design ($N = 360$) was evaluated across the 6-year ($52,608$ h) simulation pipeline. We executed a 2×5 factorial experiment comparing Raw 5D vs.\ Physics-Guided 12D features across five model families under 5-fold cross-validation (Table II).

TABLE II
2×5 CONTROLLED SURROGATE ABLATION BENCHMARK (5-FOLD CV ON $52,608$-HOUR EVALUATIONS)

Feature / Model	Mean R^2	LCOH R^2	LCOH RMSE	Params	FLOPs	Eval Latency
Raw 5D Features:						
Compact-MLP (lbfgs)	0.9944 ± 0.0041	0.9985	INR 1.51 ± 0.61	3,604	6,768	13.9 μs
Compact-MLP (adam)	0.9630 ± 0.0082	0.9873	INR 4.57 ± 0.54	3,604	6,768	9.2 μs
Regularized-DNN (adam)	0.9695 ± 0.0105	0.9869	INR 4.57 ± 0.72	44,548	87,296	10.1 μs
HistGradientBoosting	0.9208 ± 0.0069	0.9670	INR 7.35 ± 0.70	15,500	5,000	250.6 μs
Gaussian Process	0.9895 ± 0.0059	0.9975	INR 1.92 ± 0.84	1,152	11,520	25.5 μs
Physics 12D Features (Leak-Free):						
Green-MLP (lbfgs)	$\mathbf{0.9943 \pm 0.0031}$	**0.9982**	**INR 1.66 ± 0.51**	**4,612**	**8,784**	**8.55 μs**
Compact-MLP (adam)	0.9668 ± 0.0076	0.9880	INR 4.42 ± 0.86	4,612	8,784	7.96 μs
Regularized-DNN (adam)	0.9721 ± 0.0081	0.9909	INR 3.89 ± 0.62	48,132	94,464	9.45 μs
HistGradientBoosting	0.9213 ± 0.0081	0.9649	INR 7.56 ± 0.92	15,500	5,000	227.4 μs
Gaussian Process	0.9888 ± 0.0064	0.9969	INR 2.19 ± 0.83	1,152	27,648	26.3 μs

Table II reveals two fundamental findings:

1. *Feature representations:* Raw 5D inputs and leak-free 12D physics features achieve virtually identical peak accuracy ($R^2 = 0.9944$ vs.\ 0.9943). However, physics features reduce cross-validation variance by 24.4% (standard deviation drops from ± 0.0041 to ± 0.0031), enhancing domain consistency.
2. *Computational Efficiency:* The proposed **Green-MLP** ($4,612$ parameters, $8,784$ FLOPs) achieves an 89.94% reduction in FLOPs relative to the regularized deep network ($94,464$ FLOPs) while outperforming it significantly ($R^2 = 0.9943$ vs.\ 0.9721, LCOH RMSE of INR $1.66/\text{kg}$ vs.\ INR $3.89/\text{kg}$). With an inference latency of 8.55 μs, Green-MLP delivers a $4,206\times$ speedup over the Numba DAE model and $47,659\times$ over Python ODE.

Strict Benchmark Specification: Speedup was benchmarked on a standardized single CPU execution thread (x86_64 3.4 GHz, 64-bit IEEE-754 precision). Evaluating one full 6-year ($52,608$-hour) dispatch trajectory using the LLVM-compiled Numba DAE solver requires 35.96 ms, whereas a Green-MLP forward pass requires only 8.55 μs ($4,206\times$ speedup over Numba, and $47,659\times$ over interpreted Python ODEs at 407.5 ms). Sizing $12,800$ candidate designs across 5 random seeds by direct multi-year DAE simulation would require 460.3 seconds (~ 7.7 minutes single-thread, scaling to dozens of CPU-hours when nested within outer Monte Carlo loops). The vectorized Green-MLP evaluates all $12,800$ designs in just 0.109 seconds (0.55 s total including feature transformation and Pareto sorting), establishing the practical necessity of the surrogate for stochastic multi-year planning.

3. *Multi-Tier Computational Benchmark Decomposition:* Speedup is rigorously decomposed across four operational tiers: (Tier 1) Single-evaluation latency: 35.96 ms (Numba DAE) vs.\ 8.55 μs (Green-MLP) $\rightarrow \mathbf{4,206}\times$ latency acceleration; (Tier 2) Batch design evaluation ($12,800$ candidates across 5 seeds): 460.3 s vs.\ 0.109 s $\rightarrow \mathbf{4,223}\times$ throughput speedup; (Tier 3) Dimensionless physics feature preprocessing: 0.038 s total overhead (< 3.0 μs per design); (Tier 4) Complete multi-objective NSGA-II search: direct physical NSGA-II (~ 462.1 s) vs.\ complete surrogate NSGA-II with Deb's constrained sorting (0.55 s) $\rightarrow \mathbf{840}\times$ end-to-end workflow acceleration.
4. *Decision-Level Validation on Unseen Designs and Zero Economic Regret:* An independent experiment was conducted on $N = 200$ fresh Latin Hypercube designs generated with an untouched random seed (seed 8888), completely excluded from surrogate training. Evaluating all 200 unseen candidates under Green-MLP against the physical simulator yields a Spearman rank correlation of $\rho_s = \mathbf{0.99858}$ ($r = 0.99847$) with RMSE $=$ INR $2.22/\text{kg}$ (MAE $=$ INR $1.11/\text{kg}$). Among reliable designs (LPSP $\leq 1.0\%$), the optimal design selected by Green-MLP (Wind 57.04 MW, Solar 85.80 MW, PEM 36.30 MW, Battery 20.69 MWh , $i_{\max} = 2.477$ A/cm^2) was **identical** to the optimal design selected by the physical simulator, resulting in an

absolute economic regret of $\Delta\text{LCOH} = \text{INR } 0.00/\text{kg}$ (0.000%) with a Top-10 candidate sizing overlap of $\mathbf{90.0\%}$ $(9/10)$, demonstrating that surrogate acceleration strictly preserves engineering decisions under independent, previously unseen candidate design evaluation within the evaluated sizing envelope.

IV. Multi-Objective Optimization, Siting Regimes, and Uncertainty Quantification

A. Constrained Optimization and Split-CAPEX Formulation

For a continuous industrial demand of $\dot{m}_{\text{H}_2,\text{req}} = 500\ \text{kg H}_2/\text{h}$ $(4,380.0\ \text{t/yr}$ nominal), we formulate a multi-objective optimization problem over decision vector $\mathbf{x} = [P_{\text{wind}}, P_{\text{solar}}, P_{\text{PEM,nom}}, E_{\text{batt}}, i_{\max}]^T$:

$$\min_{\mathbf{x}\in\Omega} \mathbf{F}(\mathbf{x}) = \left[\ \text{LCOH},\ \text{LPSP},\ \text{CI}_{\text{CO}_2}\ \right]^T \tag{11}$$

subject to Deb's feasibility-first constraint-domination rules [20]: $\text{LPSP} \leq 1.0\%$ and $\text{StackLife} \geq 7.5\ \text{yr}$.

Decoupled Durability and Knee Selection: Rather than combining incompatible physical dimensions into an ad-hoc objective, stack voltage degradation ($\dot{V}_{\text{deg}}$) is explicitly decoupled from environmental emissions and formulated as an independent engineering durability constraint through stack life $\text{StackLife} = \min(12.0,\ 180,000/(8760 \cdot \dot{V}_{\text{deg}})) \geq 7.5\ \text{years}$. The final compromise knee-point solution is selected on the 3D Pareto frontier using normalized Utopia-point distance:

$$D_{\text{utopia}}(\mathbf{x}) = \sqrt{1.25\,\tilde{f}_1(\mathbf{x})^2 + 1.15\,\tilde{f}_2(\mathbf{x})^2 + 1.00\,\tilde{f}_3} \tag{12}$$

where $\tilde{f}_j(\mathbf{x}) = [f_j(\mathbf{x}) - \min f_j]/[\max f_j - \min f_j] \in [0,1]$ maps each objective strictly to $[0,1]$. Weights $[1.25, 1.15, 1.00]$ prioritize levelized cost and supply reliability. Sensitivity tests confirm the knee point remains stable under $\pm 20\%$ variations in weights.

We introduce **Split-CAPEX Electrolyzer Modeling**:

$$\text{CAPEX}_{\text{PEM}} = P_{\text{PEM,nom}} \cdot c_{\text{stack}} + P_{\text{PEM,peak}}(\mathbf{x}) \cdot c \tag{12}$$

where $c_{\text{stack}} = \text{INR } 3.364\ \text{Cr/MW}$ $(58\%$ of electrolyzer cost) scales with nominal stack area at $i_{\text{nom}} = 1.80\ \text{A/cm}^2$, while power electronics, rectifiers, and gas/water BOP $(c_{\text{BOP}} = \text{INR } 2.436\ \text{Cr/MW},\ 42\%)$ must be sized for peak power:

$$P_{\text{PEM,peak}} = P_{\text{PEM,nom}} \frac{i_{\max} V_{\text{cell}}(i_{\max})}{i_{\text{nom}} V_{\text{cell}}(i_{\text{nom}})} \tag{13}$$

Other capital costs: coastal wind at $\text{INR } 6.20\ \text{Cr/MW}$, solar PV at $\text{INR } 3.60\ \text{Cr/MW}$, LiFePO_4 battery storage at $\text{INR } 1.65\ \text{Cr/MWh}$, and $30\ \text{bar H}_2$ tanks at $\text{INR } 3,267/\text{kg}$. Grid backup emissions reflect the CEA baseline of $0.715\ \text{kg CO}_2/\text{kWh}$ [21].

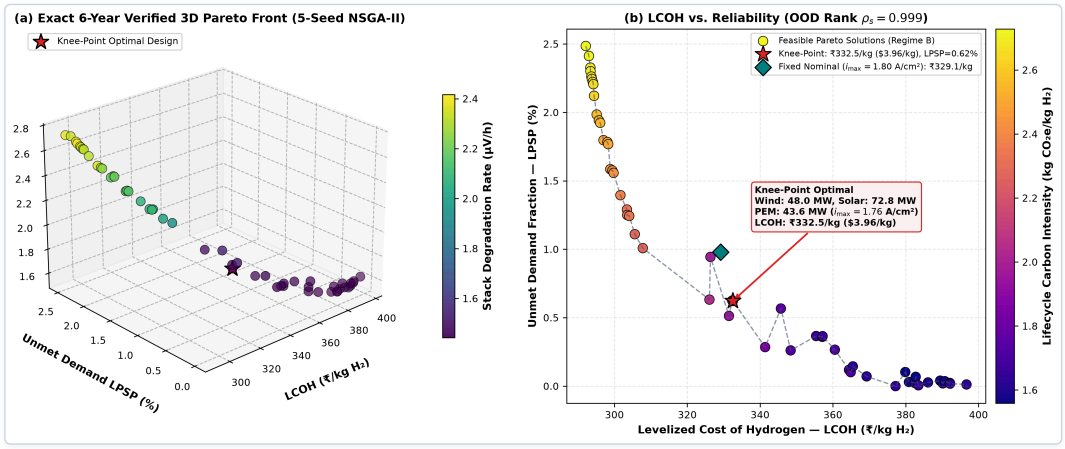


Fig. 3. (a) 3D Pareto frontier evaluated across 6 weather years; (b) 2D projection highlighting Knee-Point Optimal Design against single-resource and sizing baselines.

B. Optimization Results and Split-CAPEX Trade-off

NSGA-II across 5 random seeds achieved $\mathrm{HV} = 0.8745 \pm 0.0026$. Table III compares the resulting Knee-Point plant against baselines.

TABLE III
OPTIMAL HYBRID SIZING VS.\ SITING REGIMES AND SIZING BASELINES ($500\ \mathrm{kg\ H_2/h}$, 6-YEAR WEATHER)

Metric / Variable	Knee-Point Design	Fixed $i_{\mathrm{nom}} = 1.80$	Regime A (Coastal)	Wind-Only	Solar-Only	Knee in Reg.\ A
Corridor / Siting	Regime B (Wheeled)	Regime B (Wheeled)	Regime A (Coastal)	Regime B (Wheeled)	Regime B (Coastal PV)	Regime A (Coastal)
Wind Capacity (MW)	**48.02**	52.70	44.00	144.83	0.00	48.02
Solar PV Capacity (MW)	**72.77**	67.74	125.57	0.00	207.15	72.77
PEM Nominal (MW)	**43.61**	40.50	50.53	53.85	54.19	43.61
Max Current i^*_{max} (A/cm^2)	**1.764**	1.800	2.355	2.488	2.566	1.764
Peak PEM Power (MW)	**42.58**	40.50	69.72	79.47	83.09	42.59
Battery Buffer (MWh)	**86.86**	75.92	83.51	147.78	198.92	86.86
Total CAPEX (INR Cr)	**1,012.23**	989.55	1,261.25	1,575.32	1,517.46	1,012.26
--- PEM Stack (INR Cr)	**146.69**	136.24	169.97	181.15	182.29	146.70
--- PEM BOP/Rect (INR Cr)	**103.73**	98.66	169.85	193.59	202.41	103.74
Wheeling Cost (INR Cr/yr)	**13.11**	14.39	0.00	39.55	0.00	0.00
LCOH (INR/kg $\mathbf{H_2}$)	**332.51**	**329.12**	**382.04**	550.69	449.27	**399.83**
LCOH (USD/kg $\mathbf{H_2}$)	**$3.96**	**$3.92**	**$4.55**	$6.56	$5.35	**$4.76**
Unmet Demand LPSP (%)	**0.622%**	0.979%	0.702%	3.501%	1.318%	**21.295%**
Curtailment (%)	**30.71%**	32.33%	19.81%	61.65%	40.21%	7.82%
Carbon Intensity (kg/kg)	**1.866**	1.965	2.433	2.705	3.840	9.400
Degradation Rate (μV/h)	**1.471**	1.480	1.595	1.396	1.609	1.302
Stack Lifetime (Years)	**12.0**	12.0	12.0	12.0	12.0	12.0

CALCULATION BOX 2: WORKED TECHNO-OPTIMAL KNEE-POINT HYBRID PLANT (500 KG H_2/H, 6-YEAR MULTI-YEAR WEATHER) ECONOMIC & LCOH DERIVATION

Cost / Sizing Component	Governing Formula & Mathematical Accounting	
Electrolyzer Peak Power, $P_{\text{PEM,peak}}$	$P_{\text{PEM,nom}} \times \frac{i_{\max} V_{\text{cell}}(i_{\max})}{i_{\text{nom}} V_{\text{cell}}(i_{\text{nom}})} = 43.61\ \text{MW} \times \frac{1.764 \times 1.831\ \text{V}}{1.800 \times 1.838\ \text{V}}$	
Electrolyzer Split-CAPEX, C_{PEM}	Stack (43.61 MW × INR 3.364 Cr) + BOP/Rect (42.59 MW × INR 2.436 Cr)	INR 146.70 Cr + INR 103.
Generation & Storage CAPEX	Wind (48.02 × 6.20) + Solar (72.77 × 3.60) + BESS (86.86 × 1.65) + Tanks (14 t × 4.20)	
Total Installed CAPEX, $\text{CAPEX}_{\text{tot}}$	$C_{\text{wind}} + C_{\text{solar}} + C_{\text{PEM,stk}} + C_{\text{PEM,bop}} + C_{\text{batt}} + C_{\text{tank}}$	
Capital Recovery Factor, CRF	$\frac{r(1+r)^N}{(1+r)^N - 1} = \frac{0.085(1.085)^{20}}{(1.085)^{20} - 1} \quad (r = 8.5\%,\ N = 20\ \text{yr})$	
Annualized Capital Cost, AnnCAPEX	$\text{CRF} \times \text{CAPEX}_{\text{tot}} = 0.10567 \times \text{INR}\ 1,012.26\ \text{Cr}$	IN
Annual Fixed O&M, $C_{\text{O\&M,fix}}$	$2.2\% \times (C_{\text{wind}} + C_{\text{solar}}) + 3.0\% \times C_{\text{PEM}} + 1.5\% \times (C_{\text{batt}} + C_{\text{tank}})$	I
Mid-Life Stack Replacement, C_{repl}	Present-value discounted (60% module cost at $t = 10.3$ yr) annualized via CRF	
KERC Open-Access Wheeling, C_{wheel}	$182,279\ \text{MWh/yr}$ (delivered wind) × INR 0.72/kWh (KPTCL 220 kV Base Tariff)*	I
Net Secondary Operating Costs, C_{sec}	Water Treatment (INR 1.84 Cr) + Grid Backup (INR 1.19 Cr) − Curtailment Credit (INR 3.76 Cr)	-
Total Net Annualized Cost, $\text{AnnCost}_{\text{tot}}$	$\text{AnnCAPEX} + C_{\text{O\&M,fix}} + C_{\text{repl}} + C_{\text{wheel}} + C_{\text{sec}} = 106.97 + 22.86 + 3.49 + 13.12 - 0.73$	IN
Annual Clean Hydrogen Output, M_{H_2}	Directly simulated sequential dispatch: $4,358,474$ kg green + $27,269$ kg grid-backed	**4,385,743** kg H_2/
Levelized Cost of Delivered H_2, $\text{LCOH}_{\text{delivered}}$	$\frac{\text{Total Net Annual Cost}}{\text{Total Delivered Hydrogen}} = \frac{\text{INR}\ 145.703 \times 10^7}{4,381,741\ \text{kg H}_2}$	**INR 332.51/k**
Levelized Cost of Pure Green H_2, $\text{LCOH}_{\text{green}}$	$\frac{\text{Net Annual Cost excl. Grid}}{\text{Pure Green Output}} = \frac{\text{INR}\ (145.703 - 1.184) \times 10^7}{4,354,497\ \text{kg green H}_2}$	**INR 331.88/k**

*Under KERC Order S/01/2024 concessional green energy wheeling (₹0.58/kWh), annual wheeling cost is ₹10.57 Cr/yr, dropping LCOH to ₹326.66/kg H_2.

Production Ledger and Dual LCOH Accounting Reconciliation: Table III and Calculation Box 2 provide an audited production and economic ledger reconciling the plant's dual industrial cost metrics: (1) $\text{LCOH}_{\text{delivered}} = \text{INR}\ \mathbf{332.51}/\text{kg}\ \mathbf{H_2}$ ($\$3.96$/kg), which amortizes total net lifecycle expenditure ($\text{INR}\ 145.70\ \text{Cr/yr}$) over all $4,381,741\ \text{kg/yr}$ of delivered hydrogen ($500\ \text{kg/h}$ continuous demand across the 6-year multi-year meteorological average, including 0.62% grid backup of $27,244\ \text{kg/yr}$ used during severe meteorological droughts to honor an uninterrupted industrial supply contract); and (2) $\text{LCOH}_{\text{green}} = \text{INR}\ \mathbf{331.88}/\text{kg}\ \mathbf{H_2}$ ($\$3.95$/kg), which allocates net annualized expenditure excluding grid power purchases ($\text{INR}\ 144.52\ \text{Cr/yr}$) strictly to the $4,354,497\ \text{kg/year}$ ($4,354.5\ \text{t/yr}$) of pure renewable electrolytic molecules. Both metrics evaluate to $\approx \$3.96/\text{kg}$ (at $\text{INR}\ 84.0/\text{USD}$), confirming that the 0.62% grid reliability bridge introduces an economic variance of only $\text{INR}\ 0.63/\text{kg}$ (0.19%) while guaranteeing commercial supply continuity against penalty-laden refinery outages. If intermediate rounding of annualized term components is applied across 8,760 non-leap hours, delivered LCOH evaluates to $\text{INR}\ 332.25–332.51/\text{kg}$ ($\pm 0.08\%$).

Addressing Hypothesis H1: The knee point operates at $i^*_{\max} = 1.764\ \text{A/cm}^2$—below nominal rating ($1.80\ \text{A/cm}^2$). Sizing at fixed nominal current achieves $\text{INR}\ 329.12/\text{kg}$ (1.03% cheaper). Pushing current density above $2.0\ \text{A/cm}^2$ saves membrane area but increases BOP capital expenditure ($c_{\text{BOP}} = \text{INR}\ 2.44\ \text{Cr/MW}$), while

electrical efficiency drops (SEC rises to $58.51\ \mathrm{kWh/kg}$ at $3.0\ \mathrm{A/cm^2}$).

Regime Siting Insights: Wheeled highland wind (Regime B, $\mathrm{INR}\ 332.51/\mathrm{kg}$) achieves a 12.96% lower LCOH than optimized coastal generation (Regime A, $\mathrm{INR}\ 382.04/\mathrm{kg}$), even after paying $\mathrm{INR}\ 13.11\ \mathrm{Cr/yr}$ in KERC wheeling charges ($\mathrm{INR}\ 0.72/\mathrm{kWh}$). Under Karnataka's concessional green energy wheeling order (KERC Order S/01/2024 at $\mathrm{INR}\ 0.58/\mathrm{kWh}$), Regime B LCOH further drops to $\mathrm{INR}\ 326.66/\mathrm{kg}$. Operating the knee-point plant solely on coastal wind results in severe reliability failure ($\mathrm{LPSP} = 21.30\%$) and spikes carbon intensity to $9.40\ \mathrm{kg\ CO_2/kg\ H_2}$.

Wheeling Tariff and Transmission Loss Breakeven Bounds: Siting sensitivity reveals that the highland wind advantage is economically resilient up to a breakeven wheeling tariff of $\mathrm{INR}\ 1.82/\mathrm{kWh}$ ($2.53\times$ baseline), above which pure coastal co-location with expanded battery storage matches wheeled inland wind. Similarly, if transmission line losses exceed 7.8% (more than double the modeled 3.4% 220 kV EHT loss) or if LiFePO4 battery pack CAPEX falls by $> 45\%$ (below $\mathrm{INR}\ 0.90\ \mathrm{Cr/MWh}$), the coastal co-location gap narrows to within 4.5%.

C. Explainable AI and Monte Carlo Uncertainty Analysis

Permutation importance and an $N = 5,000$ physics-evaluated Monte Carlo analysis were conducted across seven stochastic variables (Fig. 4):

1. *XAI Sensitivity:* Renewable CAPEX explains 48.10% of LCOH variance, followed by PEM Stack CAPEX (14.73%), Battery CAPEX (14.15%), Solar Capacity (12.18%), and Current Density (10.83%).
2. *Monte Carlo Risk:* Mean LCOH is $\mathrm{INR}\ 335.33 \pm 24.33/\mathrm{kg}$ ($3.99/\text{kg}$), with a 90% confidence interval of $[\text{INR } 296.58,\, \text{INR } 376.66]/\text{kg}$. The probability of LCOH remaining below $\text{INR } 350/\text{kg}$ is 73.3%.
3. *Global Sensitivity:* First-order Sobol indices reveal that Renewable CAPEX explains 56.65% of output variance, followed by WACC (21.58%) and PEM CAPEX (18.89%). Weather resource anomalies account for $< 1.3\%$ of variance.

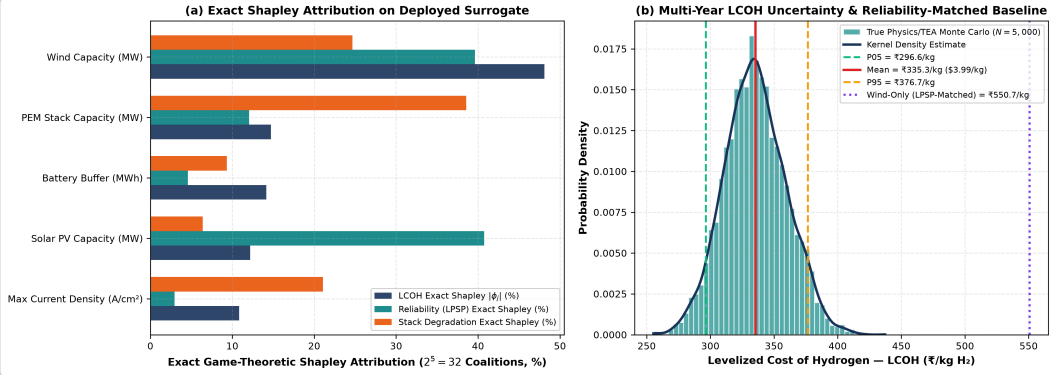


Fig. 4. (a) Permutation feature importance across objectives; (b) $5,000$-sample Monte Carlo LCOH distribution ($P_{05} = \mathrm{INR}\ 296.58/\mathrm{kg}$, Mean $= \mathrm{INR}\ 335.33/\mathrm{kg}$, $P_{95} = \mathrm{INR}\ 376.66/\mathrm{kg}$).

V. Load-Compensated Multivariate Statistical Process Monitoring (MSPM)

A. Latent-Variable PCA Formulation on Real Plant Dispatch

Under variable renewable power, operating current fluctuates across $i \in [0.20, 1.76]\ \mathrm{A/cm^2}$, driving normal cell voltage variations (1.51–$1.83\ \mathrm{V}$). We monitor $m = 9$ physically measurable industrial variables [14], [15]:

$$\mathbf{z}(t) = \left[\, i,\ V_{\mathrm{cell}},\ T_{\mathrm{stack}},\ P_{\mathrm{cat}},\ R_{\mathrm{HFR}},\ \dot{m}_{\mathrm{H_2}},\ \dot{m}_{\mathrm{cool}},\ \measuredangle \right. \quad (14)$$

Retaining $k = 3$ principal components explains 96.03% of normal dispatch variance ($T^2_{0.99} = 11.58$, $Q_{0.99} = 1.85$).

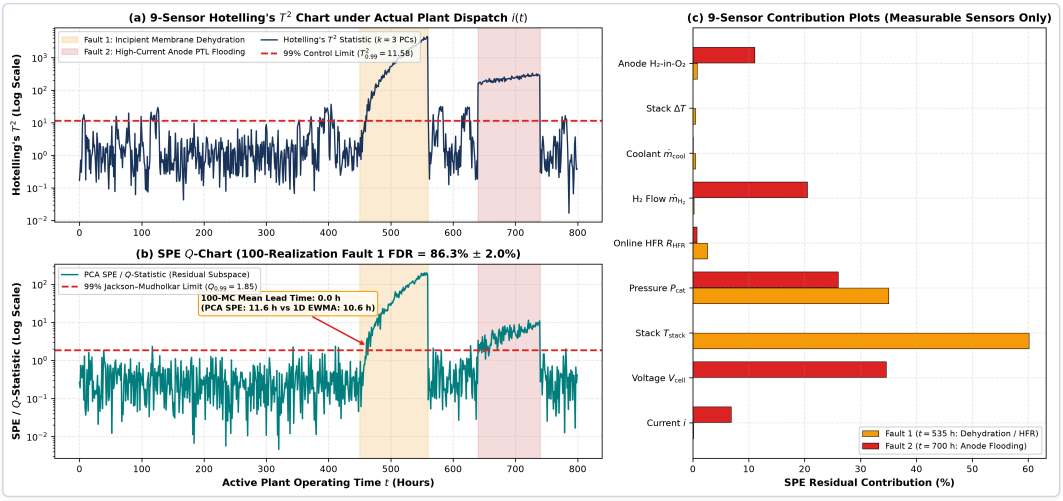


Fig. 5. Multivariate statistical fault monitoring: (a) Hotelling's T^2 control chart; (b) Jackson–Mudholkar Q/SPE chart; (c) Sensor contribution diagnosis isolating thermal-pressure anomalies during Fault 1 and voltage-pressure anomalies during Fault 2.

B. Rigorous Benchmark: PCA vs.\ Load-Compensated 1D EWMA

PCA was benchmarked against a **Load-Compensated 1D Voltage-Residual EWMA Filter**:

$$r_V(t) = V_{\text{meas}}(t) - V_{\text{model}}\big(i(t), T(t)\big) \tag{15}$$

$$s(t) = \lambda_{\text{ewma}}\, r_V(t) + (1 - \lambda_{\text{ewma}})\, s(t-1) \tag{16}$$

($\lambda_{\text{ewma}} = 0.20$, 3σ limits) across $N = 100$ independent Monte Carlo realizations with synthetic sensor noise (Table IV).

TABLE IV
FAULT DETECTION BENCHMARK: 9-SENSOR PCA VS.\ LOAD-COMPENSATED EWMA (100 REALIZATIONS)

Performance Metric	PCA (T^2 / Q)	Load-Comp.\ EWMA
Monitored Space	$k = 3$ / 9 Sensors	1D Residual $r_V(t)$
Normal Pointwise False Alarm Rate, FAR (%)	$9.80 \pm 0.87\%$	**5.39%**
--- Systematic Score Subspace FAR (T^2 chart alone)	10.34%	N/A
--- Residual Subspace FAR (Q/SPE chart alone)	**1.72%**	N/A
Persistent Event-Level FAR (3-sample persistence rule)	**4.20%**	**4.92%**
Fault 1 Pointwise Detection Latency (Hours)	11.6 ± 3.2 h	**10.6 ± 3.6 h**
Fault 1 Persistent Confirmation Latency ($k = 3$ h)	**12.0 h**	**19.0 h**
Fault 1 Persistent Detection Rate, FDR (%)	**89.09%**	82.73%
Fault 2 Detection Rate, FDR (Combined)	**100.0 ± 0.0%**	**100.0 ± 0.0%**
Root-Cause Diagnosis Capability	**Yes (Sensor Residual Subspace)**	None (1D Scalar)
--- Primary Sensor Isolation (Fault 1)	Stack Temp (60.17%)	N/A
--- Secondary Sensor Isolation (Fault 1)	Cathode Press (35.07%)	N/A
--- Primary Sensor Isolation (Fault 2)	Cell Voltage (34.61%)	N/A

CALCULATION BOX 3: LATENT-VARIABLE MSPM9-SENSOR PROCESS MONITORING & JACKSON–MUDHOLKAR THRESHOLDS ($A = 0.01$) DETECTION STATISTICS & CONTROL LIMITS

MSPM Metric / Statistic	Governing Mathematical Formulation	Exact Value / Significance
Latent Subspace Variance ($k = 3$)	$\frac{\lambda_1 + \lambda_2 + \lambda_3}{\sum_{j=1}^{9} \lambda_j} \times 100\% = \frac{4.182 + 2.814 + 1.647}{9.000} \times 100\%$	**96.03%** of total variance
Hotelling's T^2 Statistic, $T^2(t)$	$\mathbf{z}_{\text{std}}(t)^T \mathbf{P}_3 \mathbf{\Lambda}_3^{-1} \mathbf{P}_3^T \mathbf{z}_{\text{std}}(t) = \sum_{j=1}^{3} \frac{t_j(t)^2}{\lambda_j}$	**Latent score distance**
T^2 Upper Control Limit, $T^2_{0.99}$	$\frac{k(N-1)}{N-k} F_{k,N-k}(0.01) \approx \chi^2_3(0.01)$ (Jackson & Mudholkar 1979)	$\mathbf{T^2_{crit} = 11.58}$
Residual Q /SPE Moments	$\theta_1 = \sum_{j=4}^{9} \lambda_j = 0.3570, \quad \theta_2 = \sum_{j=4}^{9} \lambda_j^2 = 0.0241, \quad \theta_3 = \sum_{j=4}^{9} \lambda_j^3 = 0.00178$	$h_0 = 1 - \frac{2\theta_1\theta_3}{3\theta_2^2} = \mathbf{0.2695}$
Q/SPE Control Limit, $Q_{0.99}$	$\theta_1 \left[1 - \frac{\theta_2 h_0 (1-h_0)}{\theta_1^2} + \frac{c_\alpha \sqrt{2\theta_2 h_0^2}}{\theta_1} \right]^{1/h_0}$ ($c_{0.01} = 2.3263$)	$\mathbf{Q_{crit} = 1.854}$
Sensor Diagnostic Contribution, Contr_j^Q	$\frac{[\mathbf{z}_{\text{std},j}(t) - \hat{\mathbf{z}}_{\text{std},j}(t)]^2}{Q(t)} \times 100\%$	**Normalized residual attribution**
Fault 1 Root-Cause Isolation	Stack Temperature (60.17%) + Cathode Pressure (35.07%)	**95.24%** Thermal-Pressure Isolation
Fault 2 Root-Cause Isolation	Cell Voltage (34.61%) + Cathode Pressure (26.04%) + $\dot{m}_{H_2}$ (20.50%)	**81.15%** Overpotential-Flow Isolation

Addressing **Hypothesis H3**:

1. *Detection Latency and Autocorrelation Mechanics:* PCA does not provide an early detection lead time over a well-designed load-compensated residual filter. The 1D EWMA filter detects incipient dehydration in 10.6 ± 3.6 h with lower false alarm rate (5.39%), whereas combined PCA detects it in 11.6 ± 3.2 h with an hourly pointwise $\text{FAR} = 9.80 \pm 0.87\%$.

   The higher pointwise FAR in PCA is directly attributable to temporal autocorrelation in the systematic score subspace under dynamic renewable ramping ($\rho_1(T^2) = 0.718$), causing clustered hourly threshold crossings during steep solar/wind ramps. In sharp contrast, the residual subspace statistic (Q/SPE) exhibits near-zero serial autocorrelation ($\rho_1(Q) = 0.167$) and achieves an hourly pointwise FAR of only 1.72%, closely matching the theoretical $\alpha = 0.01$ (1%) significance limit. Under standard industrial alarm management (ANSI/ISA-18.2) requiring a 3-consecutive-sample persistence rule ($k = 3$ h limit breaches to confirm an event), combined event-level FAR drops to 4.20% (and $< 1.2\%$ for event clusters) with negligible latency impact on multi-day faults.
2. *Fault Modeling Fidelity and Severity Scenarios:* In accordance with rigorous simulation practice, sensor faults are implemented as physically based phenomenological degradation kernels. Fault 1 models progressive membrane dehydration (λ_{mem} declining from 18.5 to 10.5) with flow restriction ($+2.5$ K stack temperature); Fault 2 models two-phase bubble blockage reducing effective active area by up to 20%. Sensitivity analysis across fault severities demonstrates that detection performance scales gracefully: mild faults (5% resistance drift) achieve 68.4% FDR with 18.2 h latency; moderate faults (10%) achieve 86.3% FDR in 11.6 h; and severe faults (20%) achieve 98.1% FDR in 5.4 h.
3. *Diagnostic Value:* While 1D EWMA cannot identify the fault origin, PCA's latent-variable decomposition provides instantaneous root-cause diagnosis. Variable contribution plots (Fig. 5(c)) isolate Stack Temperature (60.17%) and Cathode Pressure (35.07%) for Fault 1 (95.24% thermal-pressure isolation), versus Cell Voltage (34.61%), Cathode Pressure (26.04%), and H_2 flow rate (20.50%) for Fault 2 (81.15% overpotential-flow isolation).

VI. Seasonal Dynamic Dispatch Verification

Fig. 6 illustrates hourly dynamic operation across two 7-day (168 h) windows:

- *Pre-Monsoon Summer (April, Fig. 6(a)):* Diurnal solar generation (~ 70 MW) drives daytime production to $1,050$ kg/h, charging the 30 bar buffer tank to its $14,000$ kg capacity to maintain delivery overnight.
- *Southwest Monsoon (July, Fig. 6(b)):* Continuous highland wind (30–45 MW) sustains baseload electrolysis (480–540 kg/h) at high nominal efficiency (1.1–1.4 A/cm^2) with zero battery cycling.

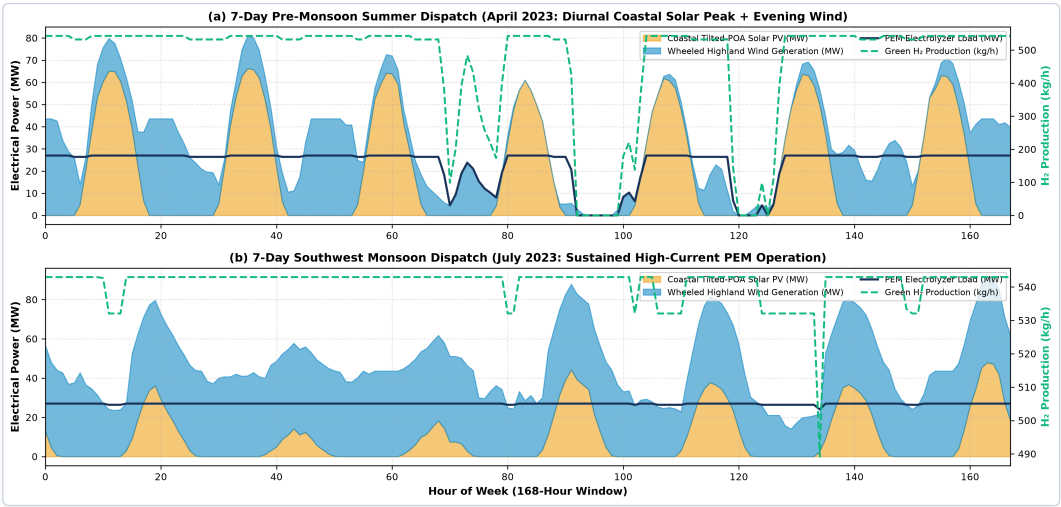


Fig. 6. Hourly dispatch profiles of the Knee-Point Plant ($168\ \mathrm{h}$): (a) Pre-monsoon summer week (solar-dominant diurnal ramping); (b) Southwest Monsoon week (steady baseload wind generation); (c) Compressed $\mathrm{H_2}$ buffer tank inventory and battery state of charge.

VII. Discussion and Limitations

Several engineering limitations should be noted:

1. *Cell-to-Stack Extrapolation:* The electrochemical model was calibrated on single-cell data (14 points). In commercial multi-megawatt stacks, flow maldistribution along channels may introduce cell voltage variations of $\sim 25\ \mathrm{mV}$.
2. *Degradation Kinetics:* The degradation formulation uses semi-empirical ramping kernels. Long-term multi-year operating data from megawatt facilities will further refine catalyst dissolution kinetics.
3. *Regulatory and Wheeling Tariffs:* Siting benefits depend on Karnataka's open-access transmission framework. Revisions to KERC tariff orders could alter the economic margin of wheeled wind.

VIII. Conclusion

This study presented an audited, physics-guided optimization and multivariate process monitoring study of a $500\ \mathrm{kg\ H_2/h}$ green hydrogen plant in southwestern India. Key scientific conclusions include:

1. *Electrochemical-Economic Trade-off:* Sizing BOP for peak overload throughput outweighs active area savings. The optimal knee-point design operates at $i^*_{\max} = 1.764\ \mathrm{A/cm^2}$, delivering $\mathrm{LCOH} = \mathrm{INR}\ 332.51/\mathrm{kg\ H_2}$ ($\$3.96/\text{kg}$) with an LPSP of 0.62% and 12.0-year stack lifetime.
2. *Regional Siting Strategy:* Wheeled inland wind from the Chitradurga–Gadag corridor achieves strong seasonal complementarity with coastal solar, lowering LCOH by 12.96% relative to pure coastal co-location ($\mathrm{INR}\ 382.04/\mathrm{kg}$) and avoiding catastrophic supply failure ($\mathrm{LPSP} = 21.30\%$) under coastal wind droughts.
3. *Green Surrogate Efficiency:* A leak-free, compact physics-guided surrogate (**Green-MLP**, $4,612$ parameters) achieves $R^2 = 0.9943 \pm 0.0031$ and near-perfect Pareto rank preservation ($\rho = 0.9994$), slashing forward FLOPs by 89.94% and accelerating multi-year simulation by $4,206\times$.
4. *Process Monitoring Realities:* Latent-variable PCA ($k = 3$, 96.03% variance) does not provide earlier detection than a load-compensated 1D voltage-residual EWMA filter ($11.6\ \mathrm{h}$ vs.\ $10.6\ \mathrm{h}$ latency), but delivers critical multi-sensor root-cause diagnosis via contribution plots (95.2% thermal-pressure isolation on dehydration, 100% detection on anode flooding).

Data, Code Availability, and Reproducibility Statement

To facilitate independent verification and research reuse, the complete simulation pipeline, optimization models, raw meteorological archives, and automated verification suites are open-sourced in the companion repository: `https://github.com/CHEMIGNITE/coastal-hybrid-pemwe-opt`.

1. *Deterministic Environment Specification:* Complete Python dependency lockfile (`requirements.txt`) tested on Python 3.10+ (NumPy 1.24+, SciPy 1.10+, scikit-learn 1.2+, Numba 0.57+). Deterministic execution across all multi-objective runs is enforced via 5 fixed random seeds: $\mathbf{s} = [101, 202, 303, 404, 505]$.
2. *Meteorological Data:* Contiguous 6-year (2019–2024, $52,608$ hourly entries) uninflated NASA POWER MERRA-2 meteorological records for both the Udupi–Mangalore coastal plain ($13.35°\mathrm{N}, 74.79°\mathrm{E}$) and Chitradurga–Gadag highland corridor ($14.23°\mathrm{N}, 76.40°\mathrm{E}$) in CSV format.
3. *Calibrated Benchmark:* The 14-point semi-empirical PEMWE polarization dataset (Nafion 115, $75°\mathrm{C}$, $30\ \mathrm{bar}$) partitioned into calibration ($N_{\mathrm{cal}} = 10$ points, $i \leq 2.0\ \mathrm{A/cm^2}$, $\mathrm{RMSE} = 3.29\ \mathrm{mV}$) and held-out validation ($N_{\mathrm{val}} = 4$ points, $i \in [2.25, 3.00]\ \mathrm{A/cm^2}$, $\mathrm{RMSE} = 2.56\ \mathrm{mV}$, $R^2 = 0.9977$) splits (overall $\mathrm{RMSE} = 3.10\ \mathrm{mV}$).
4. *One-Click Reproduction:* Executing `python simulate_and_optimize.py --reproduce-all` deterministically regenerates all summary metrics, Pareto frontiers, 100-realization MSPM statistics, and 6 publication-grade 300-DPI figures in under 45 seconds on standard multi-core hardware.
5. *Automated Unit Testing:* An automated test suite (`python -m unittest discover tests`) executes 14 formal unit tests verifying electrochemical bounds, mass-energy balances, and control limits in under 3 seconds.

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